Technical University of Crete
Wi-Fi Doctor

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1 Introduction

The goal of this project is to design and implement a Wi-Fi network performance monitoring system. By capturing and analyzing Wi-Fi packets across various setups, we are able to assess network performance and identify the underlying factors influencing its quality. We observed that either transitioning from the 2.4 GHz band to the 5 GHz band or operating in a home environment instead of an enterprise setup significantly improves Wi-Fi density. Furthermore, we identified the key factors that hinder throughput and data rate performance.

2 Design

Overview

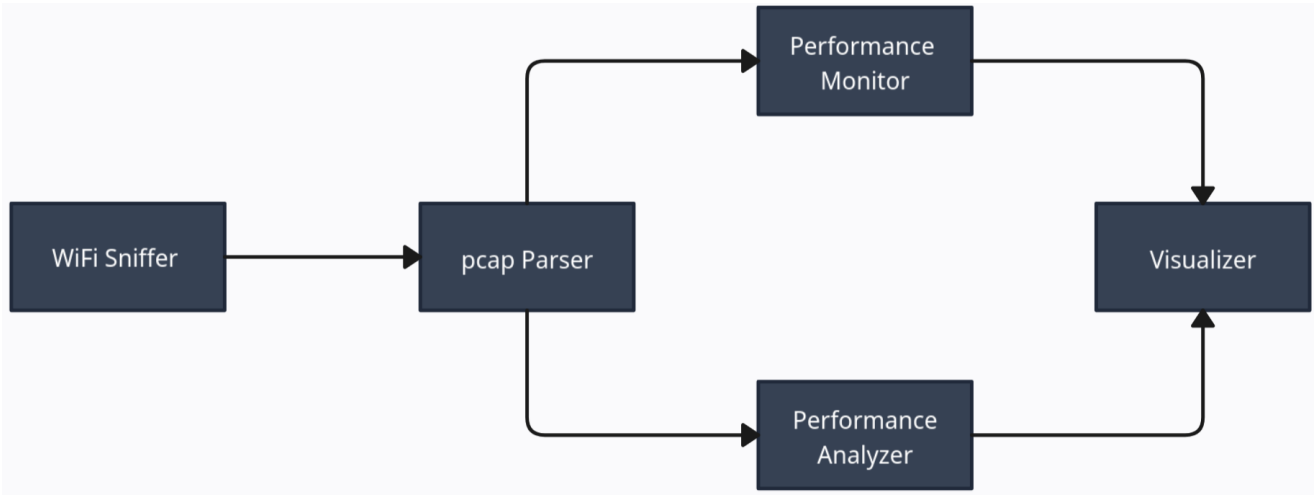


Figure 1: Pipeline diagram

1.1 Wi-Fi Network Density

Wi-Fi Sniffer

We capture PCAP files using Wireshark [1] and utilize *airodump-ng* [2] to specify the channels to be monitored. To compute *beacon jitter* [3], we monitor a single designated channel for each setup (e.g. 2.4 GHz or 5 GHz in Home and Enterprise environments) to improve measurement accuracy. In all other cases, we configure the Wi-Fi card to cycle between channels using *airodump-ng*.

pcap Parser

The packet parser is developed using PyShark [4]. It analyzes Wi-Fi packets, extracts relevant parameters (e.g. *MAC address, Channel, etc.*) and initially stores the data in a Pandas DataFrame to facilitate efficient processing of large packet volumes. The data is then converted into a CSV file for integration with the performance monitoring system.

Performance Monitor

We initially composed a set of four density metrics aimed at evaluating the level of congestion and overall quality of a wireless network environment through the analysis of beacon frame data.

The first metric, *beacon jitter* [3], is implemented via a function, which calculates the time intervals between successive beacon transmissions from the same Access Point (AP).
By comparing the observed intervals to a nominal reference value (typically 100 ms), the function derives jitter values that serve as an indicator of temporal stability. Elevated jitter levels may suggest the presence of interference or congestion. Given that beacon frames possess a carrier sensing mechanism, if the channel is busy, the beacon transmission is deferred. Consequently, persistent deviations from the nominal interval may be symptomatic of a congested or heavily utilized wireless medium.

The second metric, *RSSI-based overlap index*, captures the degree of congestion in the wireless environment by estimating the number of nearby Access Points (APs) operating on overlapping channels with sufficient signal strength to cause interference. In the 2.4 GHz band, where Wi-Fi channels significantly overlap in frequency, the metric accounts for both *co-channel interference*—when multiple APs operate on the same channel—and *adjacent-channel interference*—when APs transmit on partially overlapping channels. Interference from adjacent channels is often more disruptive than co-channel interference, as it leads to unpredictable collisions and degraded performance [5][6]. To improve the accuracy of interference estimation, the metric evaluates only APs whose received signal strength (RSSI) exceeds a defined threshold, thereby filtering out distant or weak APs that are unlikely to contribute meaningfully to congestion. By detecting both types of interference and applying a realistic signal-strength filter, the overlap index provides a more comprehensive and meaningful view of effective spectral density. In contrast, for the 5 GHz band—where channels are designed to be non-overlapping under standard conditions—the metric considers only co-channel interference. Overall, this approach offers a detailed indicator of spectrum usage and network planning efficiency across both frequency bands.

The third metric, *RSSID*[7] (Received Signal Strength Inverse Density), is intended to quantify the density and potential interference of nearby wireless networks by capturing both the number of neighboring Access Points (APs) and their relative signal strength. RSSID is computed as the sum of the inverse RSSI magnitudes of all beacon signals received from distinct APs. In this formulation, stronger signals (i.e., lower absolute RSSI values) contribute more to the overall sum, while numerous weak signals have less impact on the metric. This provides a practical means to represent environments where many weak signals exist, which may indicate high-density networks or distant interference sources. Unlike simple counts of neighboring networks, RSSID captures the signal profile of the surrounding wireless environment. A higher RSSID value suggests a dense wireless neighborhood with potentially overlapping coverage zones, which may lead to increased contention or degraded performance, even if individual signals are weak. As shown in the referenced study, while no direct correlation was found between RSSID and link-layer performance due to channel variability and external interference factors, the metric remains useful as a high-level indicator of wireless spatial density and environmental complexity.

The fourth metric, *PHY type distribution*, captures the technological composition of the observed wireless environment by analyzing the physical layer (PHY) capabilities advertised by Access Points (APs) in their beacon frames. Each AP supports a specific version of the IEEE 802.11 standard (e.g., 802.11b, g, n, ac, ax), which determines its maximum transmission rate, modulation scheme, and spectral efficiency. Environments dominated by legacy technologies such as 802.11b or 802.11g are more likely to experience congestion , as these protocols occupy airtime longer due to their lower data rates and lack of modern features like frame aggregation or multiple-input multiple-output (MIMO). In contrast, a higher proportion of modern PHY types (e.g., 802.11n, 802.11ac) typically reflects more efficient spectrum usage and higher network capacity.In dense environments, this behavior can significantly degrade overall performance, as older devices consume disproportionate amounts of spectrum resources, causing delays and collisions for newer, faster devices. Therefore, environments with a high proportion of legacy PHY types are more susceptible to congestion, especially when multiple APs and clients are competing for the same limited bandwidth. By analyzing the distribution of PHY types, this metric helps identify networks where outdated technologies may be contributing to inefficient spectrum utilization and elevated network density effects, despite a potentially low number of devices.

Visualizer

The visualizer produces four plots to assess wireless network conditions. The beacon jitter plots present time series showing irregularities in beacon transmissions, indicating potential congestion. The average overlap index plot compares how congested each scenario is, based on the presence of overlapping APs. The RSSID plot reflects the density of networks by combining their number and signal strength. Lastly, the average jitter plot summarizes the overall timing stability across the captured scenarios.

1.2 Wi-Fi Network Performance

Wi-Fi Sniffer

We conducted an experiment where we gradually moved away from the access point in order to observe the behavior of key performance metrics. To isolate the downlink data frames exchanged between the Access Point (AP) and the client device (laptop), we applied filters that select only data frames transmitted from the AP to the client. As a result, the analysis focused specifically on downlink traffic, allowing us to evaluate the effects of signal degradation and frame loss on throughput and rate adaptation under mobility conditions.

Performance Monitor

To evaluate the reliability and efficiency of downlink transmission, we introduced a function that processes packet-level data and derives key performance indicators. First, the overall frame loss rate is calculated as the proportion of retransmitted frames relative to the total number of downlink data frames[8]. This metric serves as a proxy for link reliability. The estimated throughput is then computed by scaling the nominal data rate based on the observed loss rate, reflecting the reduction in effective data delivery.

Performance Analyzer

We computed the *Rate Gap* [7], where the theoretical data rate $R_E\big(MCS_{theoretical}\big)$ is obtained as follows. First, we filter the *Probe Requests* to isolate the frames associated with the access point of interest. Next, we examine the Wi-Fi capabilities advertised by both the access point and the client to determine the highest number of spatial streams the link can support under ideal conditions. In particular, by referencing the MCS indices documented in [9], we identify the maximum supported MCS, which serves as our *theoretical MCS*. Furthermore, we retrieve the *RSSI* and *data rate* for each packet, enabling us to investigate the performance and root causes of under performance.

Visualizer

We generate time-series plots that collectively describe downlink performance. *Data Rate plot* displays the data rate in Mbps over time, while *Rate Gap plot* highlights deviations from the theoretical maximum MCS-based rate. *RSSI plot* reveals the received signal strength and thus the link's physical quality, and *Frame Losses plot* captures retransmissions. These losses directly impact *Throughput with Frame Loss plot*, which reflects the effective data delivery. Lastly, an *Aligned Comparison plot* correlates data rate changes with frame loss spikes, offering a unified perspective on how link conditions and retransmissions interact to shape overall Wi-Fi performance.

3 Evaluation

1.1 Wi-Fi Network Density

To enhance the interpretability of our results, we established a scoring system for each of the mentioned metrics. Additionally, we computed an overall density score that integrates all individual scores and visualized the results through plots.

For the individual scores :

- **Beacon Jitter** (BJ) : The average of all the beacon jitters for each setup.
- **Channel Overlap Index** (COI) : The average of all the COIs
- **RSSID** : The RSSID value for each setup
- **PHY** : To compute the PHY score, we assume that the improvement of each 802.11 generation is solely based on data rate. After mapping each generation to its corresponding data rate, we normalize the values using logarithmic scaling [10], which determines the weight w of each AP.

$$PHY_{score} = \sum_{i}^{K} w_i \cdot AP_i$$

, where K is the total number of APs in each setup.

2.4 GHz	**BJ**	**COI**	**RSSID**	**PHY**
Home	9.1	0.8	35.29	11
Enterprise	690.09	16.77	86.78	126.98

Table 1: Indexes for 2.4 GHz

5 GHz	**BJ**	**COI**	**RSSID**	**PHY**
Home	2.46	1	8.11	5.36
Enterprise	10.79	0.583	138.92	122.63

Table 2: Indexes for 5 GHz

For the overall density score we used the min-max normalization technique[10], in order to scale all the above scores to $[0, 1]$. Moreover, we made the assumption that each of the density metrics affect the total score equally. Hence we have that :

$$D = w_{BJ} \cdot BJ_{norm} + w_{COI} \cdot COI_{norm} + w_{RSSID} \cdot RSSID_{norm} + w_{PHY} \cdot PHY_{norm}$$

, where $w_{BJ} = w_{COI} = w_{RSSID} = w_{PHY} = 0.25$.

Finally we scale our density score in $[1, 100]$: $D_{1-100} = 10 \cdot (1 + 9 \cdot D)$.

	2.4 GHz	**5 GHz**
Home	16.235	10.579
Enterprise	91.03	54.467

Table 3: Density score (1 - 100).

Beacon Jitter Analysis

The beacon jitter plots illustrate the timing consistency of beacon transmissions from Access Points (APs) across different scenarios. For clarity, we visualized only the most frequently observed APs rather than all detected beacons from the Wireshark capture.

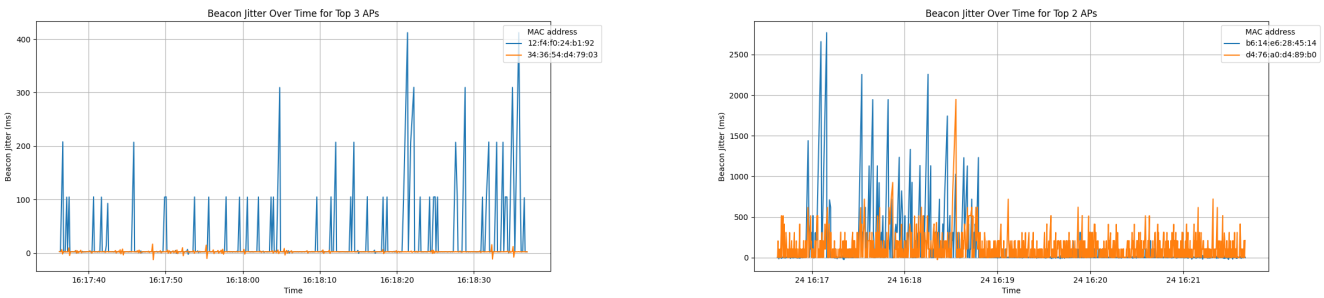
In the **2.4 GHz Home** scenario (2a), the two APs display contrasting behavior: one shows large jitter spikes up to 400 ms, likely due to missed beacons or irregular intervals, while the other maintains a stable range practically 0 ms. This highlights that even in low-traffic environments, beacon stability can vary significantly between APs.

In the **2.4 GHz Mall** scenario (2b), both APs show elevated jitter, with spikes up to 2,500 ms and frequent fluctuations in the hundreds. These variations reflect the impact of interference and congestion in a busy public environment.

The **5 GHz Home** scenario (3a) exhibits more stable behavior overall. Both APs show one spike of jitter (100ms), likely due to brief disruptions—consistent with a less congested spectrum.

In the **5 GHz Mall** environment (3b), jitter remains moderate, with one AP peaking around 200–300 ms and another fluctuating between 50–100 ms.

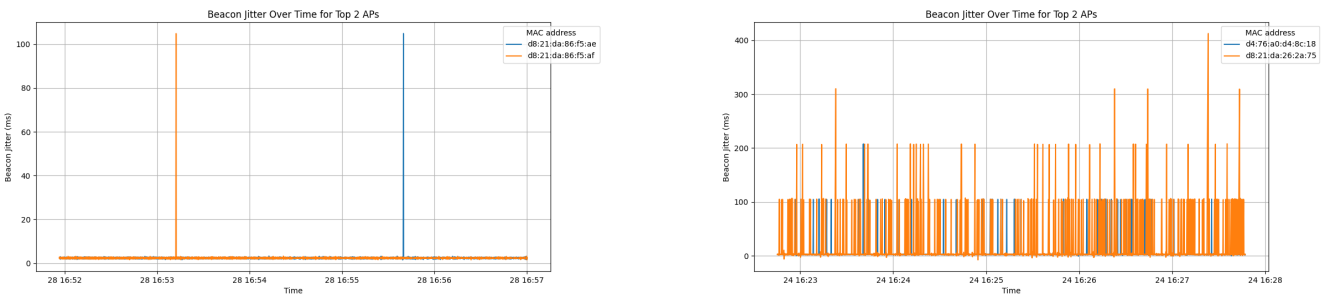
Despite occasional spikes, the 5 GHz band shows less timing variability than 2.4 GHz, indicating better performance in dense environments. These results demonstrate how both frequency band and deployment context (home vs. mall) influence beacon timing. The 2.4 GHz band, particularly in public spaces, tends to suffer from greater timing irregularities, while the 5 GHz band offers more consistent and resilient transmission patterns.



(a) 2.4 GHz Home

(b) 2.4 GHz Mall

Figure 2: Beacon jitter time series for 2.4 GHz scenarios.



(a) 5 GHz Home

(b) 5 GHz Mall

Figure 3: Beacon jitter time series for 5 GHz scenarios.

Density Metric Comparison

The plots reveal that the **2.4 GHz Mall** scenario exhibits the highest levels of wireless congestion, with peak values in both the overlap index (4) and average jitter (5). In contrast, the **5 GHz Home** environment demonstrates minimal interference and stable performance across most metrics. Notably, the **5 GHz Mall** scenario shows a higher total RSSID than the **2.4 GHz Mall** (6), which can be attributed to the capture detecting a greater number of Access Points in the 5 GHz band—likely due to broader channel coverage or denser AP deployment in that environment.

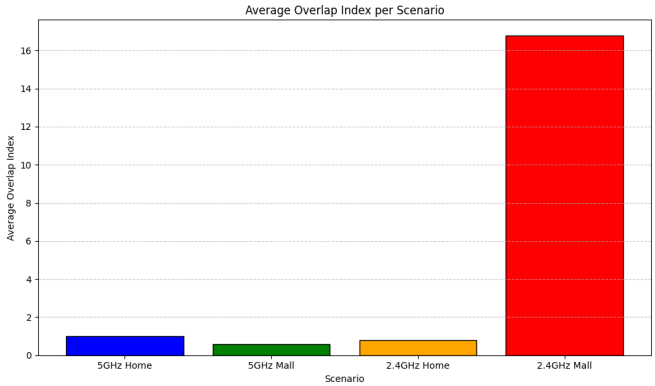


Figure 4: (a) Average Overlap Index

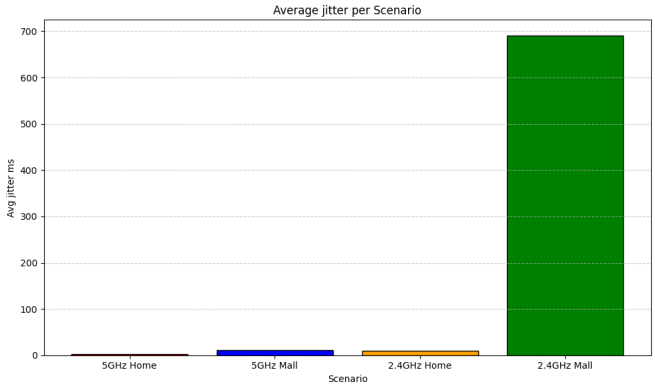


Figure 5: (b) Average Beacon Jitter

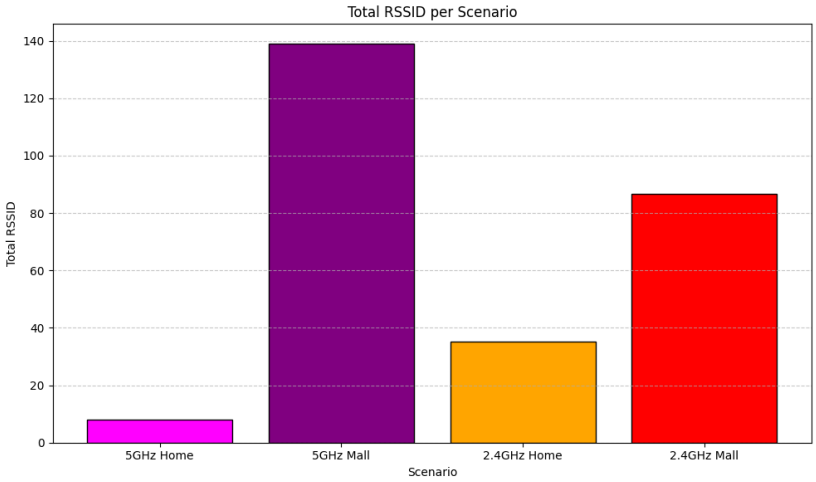


Figure 6: (c) Total RSSID per Scenario

Figure 7: Comparison of Wi-Fi density metrics: (a) overlap index, (b) jitter, and (c) RSSID

1.2 Wi-Fi Network Performance

Throughput and Data Rate Analysis

In our experiment, we processed a total of 167243 data frames, among which 27723 were retried, resulting in a frame loss rate of 16.58% and an average throughput of 99.84 Mbps.

Metric	Value
Total data frames:	167243
Retried data frames:	27723
Frame loss rate:	16.58%
Average Throughput:	99.84 Mbps

Table 4: Summary of Data Frame Metrics

- **Mobility and Signal Quality**

  As device 'A' moved away from the Access Point (AP), the received signal strength (RSSI) gradually declined (8). This degradation in signal quality led to a significant drop in the observed data rate (9) and a corresponding increase in the rate gap (10), indicating that the effective throughput diverged from the PHY layer's potential. These effects are attributed to the adaptive rate control mechanisms of the IEEE 802.11 protocol, which lower the modulation and coding schemes in response to poor channel conditions. As the device approached the AP again, RSSI improved, triggering a recovery in data rate and a reduction in the rate gap. The figures clearly illustrate how physical distance and mobility directly impact link quality and throughput efficiency in Wi-Fi networks.

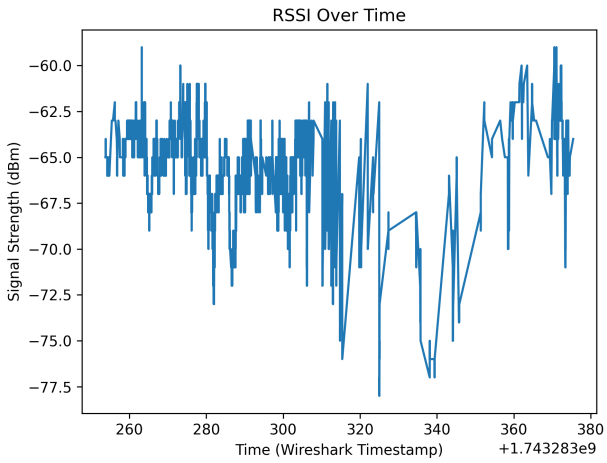


Figure 8: (a) RSSI Time series

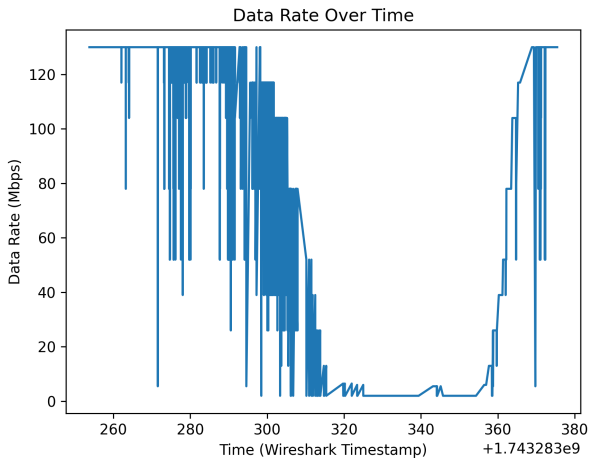


Figure 9: (b) Data Rate Time series

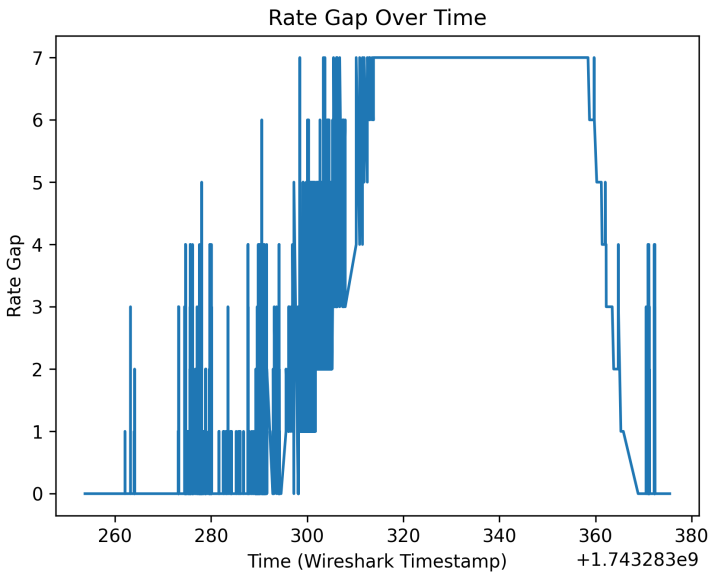


Figure 10: (c) Rate Gap Time series

Comparison of downlink values through wireshark time: (a) RSSI , (b) Data Rate , and (c) Rate Gap

- **Impact of Frame Losses on Data Rate and Throughput**

  To better understand the observed performance degradation during mobility, we analyzed the relationship between data rate, frame losses, and throughput across three plots. In 11, we observe that between timestamps approximately 290 and 310 seconds, the data rate (blue line) remains nonzero—often above 50 Mbps—indicating that the device continues to transmit frames. Simultaneously, the red line representing scaled frame losses shows dense and frequent spikes, pointing to a high number of unsuccessful transmissions during this same interval.

  In 12, we see that throughput (green line) drops significantly during the same time range, approaching zero. This reveals that despite an active link and ongoing transmissions at the PHY layer, very little data is successfully delivered—indicating that most of the transmitted frames are being lost or repeatedly retransmitted. The reason for this mismatch lies in MAC-layer behavior: when frame losses increase, the channel becomes less efficient due to retransmission overhead and contention, causing throughput to collapse even when the data rate appears acceptable.

  Finally, in 13 —which uses time bins rather than exact timestamps—shows a sharp spike in total frame losses peaking near this same interval, further supporting the finding.

  Collectively, these plots illustrate that it is not just the signal strength or rate adaptation that limits performance, but the accumulation of frame losses that can dramatically reduce usable throughput in mobile scenarios.

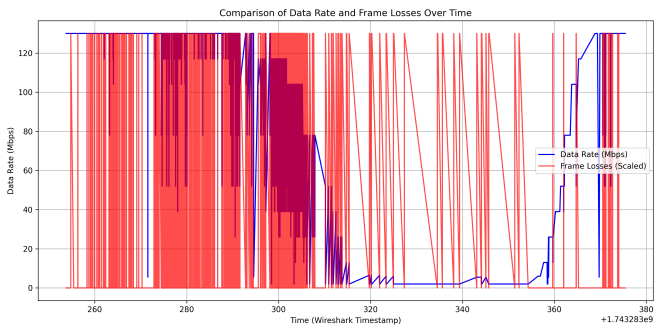


Figure 11: (a) Data Rate and frame loss Time series

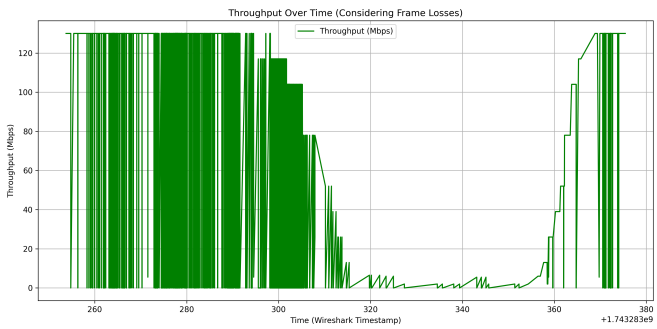


Figure 12: (b) Throughput Time series

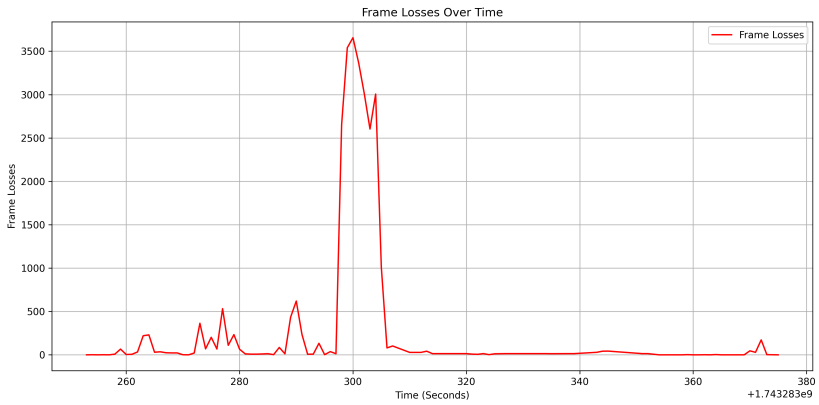


Figure 13: (c)Frame Losses Time series

Comparison of downlink values through wireshark time: (a) Data Rate and frame losses(scaled) through time, (b) Throughput over time , and (c) Frame Losses (time axis in bins)

4 Related Work

[3]

This study proposes a method to detect channel saturation in Wi-Fi networks by monitoring the jitter in IEEE 802.11 beacon transmissions. Their approach is based on the idea that when beacon intervals deviate from the expected timing, it may indicate network interference or congestion. This work has influenced our system design by providing a foundation for the beacon jitter metric, which we use to assess channel stability and performance under different network conditions.

[7]

This research article focuses on characterizing home wireless performance from the perspective of the network gateway. The study examines how factors such as interference, signal strength, and the density of nearby access points affect overall network throughput. This work provided us with Received Signal Strength Inverse Density (RSSID) metric, which quantifies the impact of neighboring access points on network performance. By considering both the number of access points and their signal strengths, the RSSID provides a more detailed understanding of wireless interference in dense environments.

References

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